ModelIt!

Student Activity Pack

Why Do Roller Coasters Make You Scream?

Modeling Forces, Energy Conversion, and the Physics of Thrills

Grade 10L1 | NGSS HS-PS2-1, HS-PS3-2 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is it about a roller coaster that makes your stomach drop or pushes you into your seat?

2 Where on a roller coaster are you going fastest? Where are you going slowest? Why?

3 Draw an energy diagram showing what happens to energy as a roller coaster goes from the top of the first hill to the bottom.

4 Why can't the second hill on a roller coaster ever be taller than the first?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Coaster Mass | Track Height | Velocity | G-Force on Rider | Energy Conversion

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Maximum Drop

Track Height: Maximum first hill | Full drop

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Mass Comparison

Same track | Different coaster masses

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Loop-the-Loop

Track includes a vertical loop

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Friction and Air Resistance — Energy losses due to wheel friction on the track and aerodynamic drag, which continuously reduce the coaster's total mechanical energy and limit how far the ride can travel

Track Curvature Radius — The radius of curves, loops, and transitions, which determines centripetal acceleration — tighter curves at high speed produce higher G-forces

Rider Body Position — How the rider's body orientation relative to the direction of force affects the G-force experience — positive Gs push blood down, negative Gs push blood to the head

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do roller coaster designers use physics to create the most thrilling — but still safe — ride possible? Use evidence from your simulation to support your answers.

1 How does your model demonstrate the law of conservation of energy on a roller coaster?

2 Why does mass cancel out of the velocity equation for an ideal coaster — and what does this tell us about gravity?

3 Where on your coaster do riders experience maximum positive G-force? Maximum negative G-force? Weightlessness?

4 How does friction change the energy conservation equation, and what does this mean for coaster design?

5 What are the limitations of modeling a roller coaster with only five components?

8 STEM Challenge: Design the Ultimate Roller Coaster

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YOUR ENGINEERING MISSION

Design a roller coaster track that maximizes rider excitement (measured by G-force variations and speed) while keeping all forces within safe human limits (under 5G sustained, under 6G instantaneous).

THE SCENARIO

A theme park has hired your engineering team to design their new signature coaster. The park has a 60-meter height limit and requires the ride to last at least 90 seconds. The coaster must include at least one vertical loop, one high-speed turn, and one airtime hill. Your design must prove — using physics calculations — that no point on the track exceeds safe G-force limits.

Engineering Design Thinking

1. WHAT HEIGHT WILL YOUR FIRST HILL BE AND HOW MUCH POTENTIAL ENERGY DOES THAT GIVE THE COASTER?
2. HOW WILL YOU CALCULATE THE MINIMUM SPEED NEEDED AT THE TOP OF EACH LOOP?
3. WHERE WILL YOU PLACE BANKED TURNS TO MANAGE LATERAL G-FORCES?
4. HOW DOES FRICTION AFFECT YOUR DESIGN — WILL THE COASTER HAVE ENOUGH ENERGY TO COMPLETE THE FULL TRACK?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Roller Coaster Engineers (a specialty within Mechanical Engineering) design rides using fluid dynamics, structural analysis, and biomechanical G-force modeling. They work for companies like Bolliger & Mabillard, Intamin, and Disney Imagineering, earning $75,000–$140,000/year. Biomechanical engineers who study human G-force tolerance earn $70,000–$130,000/year.